The gridpapers package

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v1.0.2 from 2021/03/27

All development happens at the repo: https://github.com/mcnees/LaTeX-Graph-Paper.

1 Introduction

Make your own quadrille, graph, hex, etc. paper! Uses the PGF/TikZ package for LaTeX, which should be part of any modern TeX installation. Lots of preset defaults to get started with ease, yet all colors and spacing are customizable.

There are example .tex files in the `examples` directory to help get you started with customization. Each tex file has an almost-empty body, with a `\usepackage` statement that you can customize. For example, engineer-pad.tex looks like this:

engineer-pad.tex

```
\documentclass{article}

\usepackage[pattern=majmin, colorset=engineer]{gridpapers}

\begin{document}
\thispagestyle{empty}
~
\end{document}
```

(The ~ in the body forces a non-empty body, or else latex wouldn't generate a PDF).

2 Usage

2.1 Options

Your graph paper is configured through a number of key/value options to the `\usepackage` command. Let's go through these options.

`pattern=\marg {name}`
Default: `std`

Which of the predefined patterns to use for the page or textarea background. The current list of pattern names is: `std`, `stdeight`, `majmin`, `dot`, `hex`, `hexup`, `tri`, `iso`, `lightcone`, `ruled`, `doubleruled`. We describe each of these patterns in Sec. 2.2. Patterns come with

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default page geometry (size and margins; see `geometry`), and default 'fullness' (whether they fill the page or not; see options `fullpage` and `textarea`).

`colorset=\marg {name}` Default: `std`
: Color presets. Valid color preset names are: `std`, `precocious`, `ghostly`, `brickred`, `engineer`, `plumpad`. A preset determines the `majorcolor`, `minorcolor`, and `bgcolor` all at once. But, you can start from a preset and then override some colors.

`majorcolor=\marg {color}`
: Override the preset "major" color. This can be a named color, or using the syntax from `xcolor` to mix colors together.

`minorcolor=\marg {color}`
: Override the preset "minor" color. As above.

`bgcolor=\marg {color}`
: Override the preset background color. As above.

`patternsize=\marg {length}`
: Override the preset pattern size. The meaning of this length argument is different for each pattern; see Sec. 2.2 for more.

`dotsize=\marg {length}` Default: `.7pt`
: Controls the size of the dots themselves for `pattern=dot`.

`fullpage`
: Make the pattern fill the whole page.

`textarea`
: Make the pattern fill only the text area of the document. At most one of the `fullpage` or `textarea` can be specified. If one is specified, it will override the default 'fullness' setting of the pattern.

`geometry=\marg {geometry spec}`
: Page geometry specification, using the syntax of the `geometry` package. This specification will override the pattern's default page geometry. However, if the `geometry` package was loaded before `gridpapers`, this option will be ignored.

2.2 Patterns

The current set of patterns:

`std`
: Quadrille, ten squares per inch. The `patternsize` option controls the side of a square. Default is `patternsize=0.1in`.

`stdeight`
: Quadrille, eight squares per inch. The `patternsize` option controls the side of a square. Default is `patternsize=0.125in`.

`majmin`
: Graph paper, eight squares per inch with a major grid every half-inch. The `patternsize` option controls the side of a small square (the larger squares are four times bigger). Default is `patternsize=0.125in`.

`dot`
: Grid of dots. The size of an individual dot is set by `dotsize`. The `patternsize` option controls the distance between dots. Default is `patternsize=0.1in`.

`hex`
: Grid of hexagons. The `patternsize` option controls the side length of a hexagon. Default is `patternsize=0.1666in`.

`hexup`
: Grid of hexagons, rotated 90 degrees from the default orientation. The `patternsize` option controls the side length of a hexagon. Default is `patternsize=0.1666in`.

`tri`
: Triangle grid. The `patternsize` option controls the side length of a triangle. Default is `patternsize=0.25in`.

`iso`
: Isometric grid. The `patternsize` option controls the side length of a triangle. Default is `patternsize=0.25in`.

`lightcone` A grid with light cones (45° lines) on a square grid. The `patternsize` option controls the side of a horizontal square (not the tipped squared). Default is `patternsize=0.25in`.

`ruled` Ruled page with bold lines. The `patternsize` option controls the vertical distance between lines. Default is `patternsize=0.2in`.

`doubleruled` Ruled page with bold lines alternating with light lines. The `patternsize` option controls the vertical distance between neighboring lines. Default is `patternsize=0.125in`.

3 Examples

3.1 Plenty of customization

Let's say you want to use the `tri` pattern, which by default fills the page. But you want it to fill just the textarea of an A4 page with 2cm margins, and you want the triangles to be .75cm long. Finally, you like the colors of the engineer set, but want a white background. Then you would write:

```
\usepackage[pattern=tri,
  patternsize=0.75cm,
  textarea,
  colorset=engineer,
  bgcolor=white,
  geometry={a4paper, margin=2cm}]{gridpapers}
```

3.2 Custom colors

Using named or blended custom colors is demonstrated in the example file `custom-colors.tex`:

custom-colors.tex

```
\documentclass{article}

\usepackage{xcolor}

% See the documentation of the xcolor package to learn about different
% color models for specifying colors
\definecolor{mydeepgreen}{rgb}{0.07, 0.56, 0.04}

% You can easily mix colors by using the ! syntax from xcolor.  Here
% we use it to mix 40% of our color with 60% white.
\usepackage[pattern=majmin,
  majorcolor=mydeepgreen,
  minorcolor={mydeepgreen!40}]{gridpapers}

\begin{document}
\thispagestyle{empty}
~
\end{document}
```

4 Implementation

```
\NeedsTeXFormat{LaTeX2e}[1994/06/01]
\ProvidesPackage{gridpapers}
    [2021/03/27 v1.0.2 Graph paper backgrounds]

\RequirePackage{xkeyval}
\RequirePackage{kvoptions}
\RequirePackage{xcolor}
\RequirePackage{tikz}
\usetikzlibrary{patterns.meta,calc}
\RequirePackage{tikzpagenodes}
%% everypage has been superseded -- try to use the new builtin
%% approach, but fall back to everypage-1x if needed
%% This code is roughly taken from the new everypage code
\@ifundefined{AddToHook}{%
  \IfFileExists{everypage-1x.sty}{%
    %% If everypage is new enough to complain, avoid the complaints
    \RequirePackage{everypage-1x}
  }{\RequirePackage{everypage}}
}{%
  \newcommand*{\AddEverypageHook}[1]{%
  \AddToHook{shipout/background}{\put(1in,-1in){#1}}}
}
\RequirePackage{pagecolor}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Option parsing
%% Declare switches for processing the options.

\newif\ifGP@geometrypreviouslyloaded
\newif\ifGP@fullnessset
\newif\ifGP@fullpage
\newif\ifGP@textarea
\GP@geometrypreviouslyloadedfalse
\GP@fullnesssetfalse
\GP@fullpagefalse
\GP@textareafalse

\SetupKeyvalOptions{%
  family=GP,%
  prefix=GPOpt@%
}

\DeclareStringOption[std]{pattern}
\DeclareStringOption[std]{colorset}

\DeclareStringOption{majorcolor}
\DeclareStringOption{minorcolor}
\DeclareStringOption{bgcolor}

\DeclareStringOption{patternsize}
\DeclareStringOption[.7pt]{dotsize}

\DeclareVoidOption{fullpage}{\GP@fullpagetrue}
\DeclareVoidOption{textarea}{\GP@textareatrue}

\DeclareStringOption{geometry}

```

```
\ProcessKeyvalOptions*

%% Can only have one of fullpage or textarea
\ifGP@fullpage
  \ifGP@textarea
    \PackageError{gridpapers}{%
      Can not specify both fullpage and textarea, please remove one option}{}
    \fi
  \GP@fullnesssettrue
\fi

\ifGP@textarea
  \GP@fullnesssettrue
\fi

%% We keep track of this to know whether or not we would be overriding
%% a previously-set page geometry
\@ifpackageloaded{geometry}
  {\GP@geometrypreviouslyloadedtrue}
  {\GP@geometrypreviouslyloadedfalse%
    \PassOptionsToPackage{\GPOpt@geometry}{geometry}%
    \RequirePackage{geometry}%
  }

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Actual package code
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Some nice colors.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\definecolor{plum}{rgb}{0.36078, 0.20784, 0.4}
\definecolor{chameleon}{rgb}{0.30588, 0.60392, 0.023529}
\definecolor{cornflower}{rgb}{0.12549, 0.29020, 0.52941}
\definecolor{scarlet}{rgb}{0.8, 0, 0}
\definecolor{brick}{rgb}{0.64314, 0, 0}
\definecolor{sunrise}{rgb}{0.80784, 0.36078, 0}
\definecolor{rosiebg}{RGB}{250,247,232}
\definecolor{rosiegrid}{RGB}{186,137,113}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% The color to use for the null directions when drawing lightcones.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\colorlet{lightlines}{scarlet!30}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Pre-defined Color schemes
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Here are some pre-defined color schemes for the paper background
%% and the major and minor grid lines.  These are switched by using
%% the option colorset=<name>.  The allowed values for colorset are in
%% the list below.
\define@choicekey*{GP}{colorset}[\val\nr]%
  %% Allowed values for colorset:
  {std,precocious,ghostly,brickred,engineer,plumpad}[std]{%
  \ifcase\nr\relax
    %% std
    \colorlet{minorcolor}{cornflower!30}
    \colorlet{majorcolor}{cornflower!50}
```

```
    \colorlet{bgcolor}{white}
  \or
    %% precocious
    \colorlet{minorcolor}{rosiegrid!50}
    \colorlet{majorcolor}{rosiegrid}
    \colorlet{bgcolor}{rosiebg}
  \or
    %% ghostly
    \colorlet{minorcolor}{gray!15}
    \colorlet{majorcolor}{gray!20}
    \colorlet{bgcolor}{white}
  \or
    %% brickred
    \colorlet{minorcolor}{brick!35}
    \colorlet{majorcolor}{brick!60}
    \colorlet{bgcolor}{scarlet!8}
  \or
    %% engineer
    \colorlet{minorcolor}{chameleon!50}
    \colorlet{majorcolor}{chameleon!80}
    \colorlet{bgcolor}{chameleon!10}
  \or
    %% plumpad
    \colorlet{minorcolor}{cornflower!40}
    \colorlet{majorcolor}{cornflower!70}
    \colorlet{bgcolor}{plum!10}
  \fi
}

%% Get the specified color set from the options
\def\@setkeyhelper#1#2{%
  \setkeys{GP}{#2=#1}
}
\expandafter\@setkeyhelper\expandafter{\GPOpt@colorset}{colorset}

%% If the user further specified majorcolor, minorcolor, and/or
%% bgcolor, we now override the selected colorset
\ifx\GPOpt@majorcolor\@empty
\else
  \colorlet{majorcolor}{\GPOpt@majorcolor}
\fi
\ifx\GPOpt@minorcolor\@empty
\else
  \colorlet{minorcolor}{\GPOpt@minorcolor}
\fi
\ifx\GPOpt@bgcolor\@empty
\else
  \colorlet{bgcolor}{\GPOpt@bgcolor}
\fi

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% The size parameter -- different meanings for different patterns
%% Will be reset by pattern code
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\newcommand{\GP@patternsize}{0.1in}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% This section sets up a routine for filling a shape with
%% hexagons. Uses code from:
```

```
%% http://tex.stackexchange.com/questions/6019/drawing-hexagons/6128#6128
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% We have to delay this definition until after \GP@patternsize is
%% redefined (by the pattern selection and/or user override)
\newcommand{\GP@declarehexpat}{
\tikzdeclarepattern{
  name=hexagons,
  type=uncolored,
  bounding box={(0,0) and (3*\GP@patternsize,0.866025*2*\GP@patternsize)},
  tile size={(3*\GP@patternsize,0.866025*2*\GP@patternsize)},
  parameters={\tikzhexrotate},
  tile transformation={rotate=\tikzhexrotate},
  defaults={
    rotate/.store in=\tikzhexrotate,rotate=0,
  },
  code={
      \pgfsetlinewidth{0.6pt}
      \pgftransformshift{\pgfpoint{0mm}{0.866025*\GP@patternsize}}
      \pgfpathmoveto{\pgfpoint{0mm}{0mm}}
      \pgfpathlineto{\pgfpoint{0.5*\GP@patternsize}{0mm}}
      \pgfpathlineto{\pgfpoint{\GP@patternsize}{-0.866025*\GP@patternsize}}
      \pgfpathlineto{\pgfpoint{2*\GP@patternsize}{-0.866025*\GP@patternsize}}
      \pgfpathlineto{\pgfpoint{2.5*\GP@patternsize}{0mm}}
      \pgfpathlineto{\pgfpoint{3*\GP@patternsize}{0mm}}
      \pgfpathmoveto{\pgfpoint{0.5*\GP@patternsize}{0mm}}
      \pgfpathlineto{\pgfpoint{\GP@patternsize}{0.866025*\GP@patternsize}}
      \pgfpathlineto{\pgfpoint{2*\GP@patternsize}{0.866025*\GP@patternsize}}
      \pgfpathlineto{\pgfpoint{2.5*\GP@patternsize}{0mm}}
      \pgfusepath{stroke}
    }
  }
}


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% This section sets up a routine for filling a shape with
%% triangles.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% We have to delay this definition until after \GP@patternsize is
%% redefined (by the pattern selection and/or user override)
\newcommand{\GP@declaretripat}{
\tikzdeclarepattern{
  name=triangles,
  type=uncolored,
  bounding box={(0,0) and (\GP@patternsize,2*0.866025*\GP@patternsize)},
  tile size={(\GP@patternsize,2*0.866025*\GP@patternsize)},
  parameters={\tikztrirotate},
  tile transformation={rotate=\tikztrirotate},
  defaults={
    rotate/.store in=\tikztrirotate,rotate=0,
    },
  code={
        \pgfsetlinewidth{0.6pt}
        \pgfpathmoveto{\pgfpoint{0mm}{0mm}}
        \pgfpathlineto{\pgfpoint{\GP@patternsize}{2*0.8660254*\GP@patternsize}}
        \pgfpathlineto{\pgfpoint{0mm}{2*0.8660254*\GP@patternsize}}
        \pgfpathmoveto{\pgfpoint{0mm}{0.8660254*\GP@patternsize}}
```

```
        \pgfpathlineto{\pgfpoint{\GP@patternsize}{0.8660254*\GP@patternsize}}
        \pgfpathmoveto{\pgfpoint{0mm}{2*0.8660254*\GP@patternsize}}
        \pgfpathlineto{\pgfpoint{\GP@patternsize}{0mm}}
        \pgfpathlineto{\pgfpoint{0mm}{0mm}}
        \pgfusepath{stroke}
    }
  }
}

%%%
```

```
%% Stein (@duetosymmetry on Twitter).
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% We have to delay this definition until after \GP@patternsize is
%% redefined (by the pattern selection and/or user override)
\newcommand{\GP@declaredotpat}{
\pgfdeclarepatternformonly
  {dotgrid}%% name
  {\pgfpoint{-0.5*\GP@patternsize}{-0.5*\GP@patternsize}}%% lower left
  {\pgfpoint{0.5*\GP@patternsize}{0.5*\GP@patternsize}}%%  upper right
  {\pgfpoint{\GP@patternsize}{\GP@patternsize}}%%  tile size
  {%% shape description
    \pgfpathcircle{\pgfqpoint{0pt}{0pt}}{\GP0pt@dotsize}
    \pgfusepath{fill}
  }
}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Begin pattern execution infrastructure
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% This inner code will be set by the choicekey pattern=...
\newcommand{\GP@innerpatterncode}{}
%% This is the "outer" code to hook into every page
\newcommand{\GP@patterncode}{% No blank lines in this code!
\begin{tikzpicture}[remember picture, overlay]
%%
%% Change "thin" to "very thin" if the lines are too thick.
\tikzset{
  minorgrid/.style={minorcolor, thin},
  majorgrid/.style={majorcolor, thin},
}
\ifGP@fullpage%
\coordinate (a) at (current page.south west);
\coordinate (b) at (current page.north east);
\else%
\coordinate (a) at (current page text area.south west);
\coordinate (b) at (current page text area.north east);
\fi
%%
\GP@innerpatterncode%
%%
\end{tikzpicture}
}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Begin pattern definition code
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

\define@boolkey{GP}{patterndefaultfullness}{}
\newcommand{\GP@patterndefaultgeometry}{}
\newcommand{\GP@patterndefaultsize}{}

%% Pattern-definer-helper
%% The interface is:
%% \GP@setpattern
%%   {<true for default fullpage, false for default textarea>}
%%   {<default geometry config>}
%%   {<default pattern size>}  %% NOTE, not tile length
%%   {<contents of inner pattern code>}
```

```
\newcommand{\GP@setpattern}[4]{%
\setkeys{GP}{patterndefaultfullness=#1}
\renewcommand{\GP@patterndefaultgeometry}{#2}
\renewcommand{\GP@patterndefaultsize}{#3}
\renewcommand{\GP@innerpatterncode}{#4}
}

\define@choicekey*{GP}{pattern}[\val\nr]%
  %% Allowed values for pattern:
  {std,stdeight,majmin,dot,hex,hexup,tri,iso,lightcone,ruled,doubleruled}{%
  \ifcase\nr\relax
    %% std
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Quadrille, ten squares per inch.
%% %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{false}{letterpaper, margin=0.2in}{0.1in}{%
%% Draw a grid with 10 squares per inch.
\draw[style=minorgrid, shift={(a)}] (0,0) grid [step=\GP@patternsize] (b);
%%
%% Draw a frame around the grid.
\draw[style=majorgrid] (a) rectangle (b);
    }
  \or
    %% stdeight
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Quadrille, eight squares per inch.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{false}{letterpaper, margin=0.1875in}{0.125in}{%
%% Draw a grid with 10 squares per inch.
\draw[style=minorgrid, shift={(a)}] (0,0) grid [step=\GP@patternsize] (b);
%%
%% Draw a frame around the grid.
\draw[style=majorgrid] (a) rectangle (b);
    }
  \or
    %% majmin
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Graph paper, eight squares per inch with a major grid
%% every half-inch.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{false}{letterpaper, margin=0.25in}{0.125in}{%
%% Draw a grid with 10 squares per inch.
\draw[style=minorgrid, shift={(a)}] (0,0) grid [step=\GP@patternsize] (b);
%%
\draw[style=majorgrid, shift={(a)}] (0,0) grid [step=4*\GP@patternsize]   (b);
%%
%% Draw a frame around the grid.
\draw[style=majorgrid] (a) rectangle (b);
    }
  \or
    %% dot
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Dot grid
%% Slightly modified version of code added by Leo
%% Stein (@duetosymmetry).
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{true}{}{0.1in}{%
    \fill [pattern=dotgrid,pattern color=minorcolor] (a) rectangle (b);
    }
```

```
  \or
    %% hex
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Hex grid
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{true}{}{0.1666in}{%
    \fill [pattern=hexagons,pattern color=minorcolor] (a) rectangle (b);
    }
  \or
    %% hexup
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Hex-up grid
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{true}{}{0.1666in}{%
    \fill [pattern={hexagons[rotate=90]},pattern color=minorcolor] (a) rectangle (b);
    }
  \or
    %% tri
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Triangle grid, adjust triangle size in the preamble
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{true}{}{0.25in}{%
      \fill [pattern=triangles,pattern color=minorcolor] (a) rectangle (b);
    }
  \or
    %% iso
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Isometric grid
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{true}{}{0.25in}{%
      \fill [pattern={triangles[rotate=90]}, pattern color=minorcolor] (a) rectangle (b);
    }
  \or
    %% lightcone
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% A grid with light cones.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{false}{letterpaper, margin=.125in}{0.25in}{%
%% Draw a grid with 4 squares per inch.
\draw[style=minorgrid, shift={(a)}] (0,0) coordinate grid [step=\GP@patternsize] (b);
%%
%% Draw a border around the grid.
\draw[style=majorgrid, pattern={lightcones[myshift={(a)}]}, pattern color=lightlines] (a) rectang
    }
  \or
    %% ruled
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Ruled page with bold lines every 0.2in or 0.25in
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    \GP@setpattern{false}{letterpaper, body={8in,10.8in}}{0.2in}{%
%% Draw a ruled page with lines every 0.2in
\draw[style=majorgrid, shift={(a)}] (0,0) grid [ystep=\GP@patternsize, xstep=\paperwidth] (b);
%% Draw a frame around the grid.
\draw[style=majorgrid] (a) rectangle (b);
    }
  \or
    %% doubleruled
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Ruled page with bold lines every 0.25in and light lines
```

```
%% every 0.125 in.
%%%
```

```
\AtBeginDocument{\pagecolor{bgcolor}}
%% Actually hook it in!
\AddEverypageHook{%
\GP@patterncode%
}
```

4.1 Selective grid activation

By default, the grid pattern is applied to every page of the document. The following commands allow selective activation, which is useful when including blank note pages in a larger document alongside regular text pages.

`\textbackslash gridpaperon` Activates the grid pattern from the current page onwards.

`\textbackslash gridpaperoff` Deactivates the grid pattern from the current page onwards. This is the default state.

`\textbackslash gridpages\marg {n}` Inserts ⟨*n*⟩ blank pages with the grid pattern. The grid is automatically activated before the first page and deactivated after the last one, so surrounding pages are not affected.

Example usage:

```
Some regular text on a plain page.
\gridpages{3}
More regular text, back to plain pages.
```

```
% ---------------------------------------------------------------
% Selective grid activation
% \gridpaperon  -> activate grid from current page onwards
% \gridpaperoff -> deactivate grid (this is the default)
% \gridpages{N} -> insert N blank grid pages, then deactivate
% ----------------------------------------------------------------
\newif\ifGP@active
\GP@activefalse

\newcommand{\gridpaperon}{\GP@activetrue}
\newcommand{\gridpaperoff}{\GP@activefalse}

\newcommand{\gridpages}[1]{%
  \clearpage%
  \GP@activetrue%
  \@tempcnta=0\relax%
  \loop
    \thispagestyle{empty}%
    ~%
    \advance\@tempcnta by 1\relax%
    \ifnum\@tempcnta<#1\relax%
      \clearpage%
    \fi%
  \ifnum\@tempcnta<#1\repeat%
  \clearpage%
  \GP@activefalse%
}

\endinput

\endinput
```

Change History

v1.0.0
General: Converted to DTX file 1
v1.0.1
General: Hotfix: old installs don't have everypage-1x, use everypage 1
v1.0.2
General: Replace triangle and hexagon code to allow rotated grids, add 'ghostly' colorset 1
v1.1.0
General: Add selective grid activation: `\gridpaperon`, `\gridpaperoff`, `\gridpages` 1